Open-Weight Decision Models Under 500M Parameters

Research shortlist for lightweight on-device structured decisions, with emphasis on image analysis. Counts and availability reflect the previous web review and should be checked against exact model cards before deployment.

Executive summary

Primary multimodal candidate: Laya Vision 201M, an experimental image-and-text decision model based on the SmolVLM family. It is the clearest direct match identified in the review for image-based typed decisions below 500M parameters. Treat it as experimental, not production-proven.

Important distinction: most other small decision models listed here are text-only. A text-only model cannot directly inspect images unless paired with a separate vision model. An ONNX export does not by itself prove that the vision encoder and image preprocessing are included.

1. Multimodal candidates for image analysis

Model	Approx. parameters	Modalities / notes	Links
Laya Vision	201M	Image + optional text. Experimental SmolVLM-based fork fine-tuned for typed decisions (choice, yes/no, score).	https://huggingface.co/thaitea/laya-vision-201m https://github.com/r33drichards/laya-vision
Earlier Laya Vision / SmolVLM variant	237M	Image + text. Earlier experimental variant; verify exact checkpoint revision and task performance.	https://huggingface.co/thaitea/laya-vision https://huggingface.co/thaitea/laya-vision-smolvlm-256m-score

Recommendation: start with Laya Vision 201M for image-based typed decisions, then benchmark the earlier variant if accessible. Validate against your own images and category rules.

2. Other open-weight decision models below 500M

Model	Params	Image support	ONNX status / notes	Repository
Laya	421M	No — text-only	Found in prior review	https://huggingface.co/convaiinnovations/laya
Laya Typed-Decisions	421M	No — text-only	Found	https://huggingface.co/onnx-community/laya-typed-decisions-ONNX
Laya Multilingual	322M	No — text-only	Found	https://huggingface.co/ollaya-dev/laya
Open-Jev DeBERTa-v3-large	434M	No — text-only	Not found in review	https://huggingface.co/com-kotobalabs/open-jev-deberta-v3-large
DecisionTune 1.0	395M	No — text-only	Available per prior review	https://huggingface.co/decision-tune/decisiontune-1.0
arc-decide 0.4b-v0.2	407M	No — text-only	Not confirmed	https://huggingface.co/Arc53/arc-decide-0.4b-v0.2
Quyet-1.0-Small	328M	No — text-only	Not confirmed	https://huggingface.co/chinhnc/Quyet-1.0-Small
Dev 0.4b	399M	No — text-only	Not confirmed	https://huggingface.co/mpnikhil/dev-0.4b
Phocinae-Largha-150M-v1	144.3M	No — text-only	Not confirmed	https://huggingface.co/Phocinae/Phocinae-Largha-150M-v1
OpenDecider Nano	~400M	No — text-only	Verify exact checkpoint	https://huggingface.co/collections/manjunathshiva/opendecider
Raya	~300M	No — text-only	Not confirmed	https://huggingface.co/TextCortex/raya

3. Additional candidates to verify

Model / family	Params	Why investigate	Verification status
Gemma 3 270M Typed-Decisions	270M	Potential small structured-decision fine-tune	Exact checkpoint and license require verification
LFM2.5-350M-RLCD	~354M	Small typed-decision candidate	Exact repository and modalities require verification

4. Suggested evaluation plan

- **Task parity:** use the same image, context, candidate options and inclusion/exclusion rules for every model.
- **Decision quality:** measure accuracy, per-class precision/recall, Brier score, calibration error and sensitivity to option ordering.
- **Deployment:** measure cold-start latency, warm p50/p95 latency, throughput, peak RAM, model download size and energy per decision.

• **ONNX validation:** verify vision components, image preprocessing, model revision, license, opset, output parity and target execution provider.

• **Production gate:** calibrate any confidence threshold on held-out labeled data; consider abstaining or escalating low-confidence cases.

5. Important caveats

• The list reflects the previous web review; not every repository was independently re-verified during PDF creation.

• “Not confirmed” or “not found” means the prior review did not establish the fact; it does not prove the resource does not exist.

• Parameter count is not the same as download size or peak RAM requirement.

• Open weights do not automatically mean unrestricted commercial use. Check the exact license and any access terms.

• A community ONNX conversion may be incomplete, unmaintained or incompatible with a target mobile NPU.